

famous vector fields: gravitational force
between object 1 and object 2, depending
on where object 2 is
electric force

gradient vector field: given $f: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$
s.t. $f_x(x,y)$ and $f_y(x,y)$ exist $\forall (x,y) \in D$,
 $\nabla f(x,y) = (f_x(x,y), f_y(x,y))$ gives a 2D
vector field!

sim: $f: D \subset \mathbb{R}^3 \rightarrow \mathbb{R}$, $\nabla f(x,y,z) =$
 $(f_x(x,y,z), f_y(x,y,z), f_z(x,y,z))$

in both cases ∇f is called a gradient vector field.

we say $\vec{F} : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a gradient vector field if there exists some scalar $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}$, diff'able, s.t.

$$\vec{F}(x_1, \dots, x_n) = \nabla f(x_1, \dots, x_n) \quad \forall (x_1, \dots, x_n) \in D$$

such a vector field (for which f exists s.t. $\vec{F} = \nabla f$) is called a conservative vector field. The scalar f s.t. $\nabla f = \vec{F}$ is the potential of the vector field $\vec{F}$.

for a given conservative vector field, there exist infinitely many potentials.

e.g. show $\vec{F}(x,y)$ is conservative, i.e.

$$\vec{F}(x,y) = \left(-\frac{2x}{(1+x^2)^2}, \frac{1}{(1+x^2)} \right)$$

find f s.t. $\vec{F}(x,y) = \nabla f(x,y) = (f_x(x,y), f_y(x,y))$

$$\text{so } f_x = \frac{-2x}{(1+x^2)^2}$$

$$\text{From } f_y: \quad f_y = \frac{1}{1+x^2} \quad \text{What's } f?$$

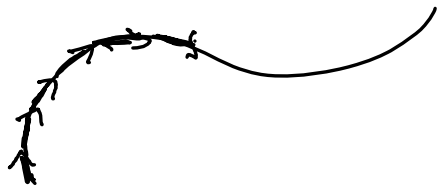
$$f(x,y) = \frac{y}{1+x^2} + g(x). \quad \text{so}$$

$$f_x = \frac{-2xy}{(1+x^2)^2} + g'(x) \quad \text{so } g'(x) = 0 \Rightarrow g(x) = C \in \mathbb{R}$$

$$\text{so } \boxed{f(x,y) = \frac{y}{1+x^2} + C} \quad \forall (x,y) \in \mathbb{R}^2$$

Now, what if I want to integrate in a vector field? along a particular path

Recall work: $\vec{F} \cdot \overrightarrow{P_0 P_1}$ if we "push an object from P_0 to P_1 ." Now, what if we push along a curve, not necessarily a straight line?



Let $\vec{F}: D \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a cont. vector field and let C be a curve described by the vector $\vec{r}: [a, b] \rightarrow \mathbb{R}^n$ s.t. $\vec{r}(t) \in D \forall t \in [a, b]$. Then the line integral of the vector field $\vec{F}$ along the curve C is defined as

$$\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt$$

since $\vec{r}'(t) = \frac{d\vec{r}}{dt}$

e.g. $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F}(x, y, z) = (x, -y^2, xz)$

and C is the curve described by the vector $\vec{r}(t) = (-t^2, 2t+1, t^2-1)$ $-1 \leq t \leq 1$
 $\vec{r}'(t) = (-2t, 2, 2t)$

$$\vec{F}(r_1(t), r_2(t), r_3(t)) = (-t^2, -(2t+1)^2, -t^2(t^2-1))$$

$$= (-t^2, -(4t^2+4t+1), -t^4+t^2)$$

$$\int_{-1}^1 -t^2 \cdot -2t - 2(4t^2+4t+1) - 2t^5 + 2t^3 dt$$

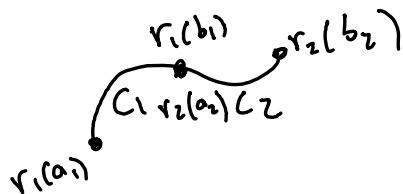
$$= \int_{-1}^1 2t^3 - 8t^2 - 8t - 2 - 2t^5 + 2t^3 dt$$

boom.

Properties: Let $\vec{F}_1, \vec{F}_2 : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ be continuous vector fields. Let C be a curve contained in D . Then

$$a) \int_C (\alpha \vec{F}_1 + \beta \vec{F}_2) \cdot d\vec{r} = \alpha \int_C \vec{F}_1 \cdot d\vec{r} + \beta \int_C \vec{F}_2 \cdot d\vec{r}$$

b) $\int_C \vec{F} \cdot d\vec{r} = \int_{C_1} \vec{F} \cdot d\vec{r} + \int_{C_2} \vec{F} \cdot d\vec{r}$, where the curves C_1 and C_2 are curves in D s.t. $C = C_1 \cup C_2$ and if C_i is described by the vector fn $\vec{r}_i : [a_i, b_i] \rightarrow \mathbb{R}^n$ then $\vec{r}_1(b_1) = \vec{r}_2(a_2)$



Fundamental thm of calc for line integrals
Let $f: D \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuously differentiable
 f^x and let C be a regular curve contained
in D and described by the vector f^n
 $\vec{r}: [a, b] \rightarrow \mathbb{R}^n$. Then

$$\int_C \nabla f \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$$

final-initial

so if $\vec{F}$ is a conservative vector field
with $\vec{F} = \nabla f$ then

$$\int_C \vec{F} \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$$

Def: path independence Let $F: D \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a continuous vector field. We say that the line integral $\int_C \vec{F} \cdot d\vec{r}$ is independent of path if $\int_{C_1} \vec{F} \cdot d\vec{r} = \int_{C_2} \vec{F} \cdot d\vec{r}$ for any 2 curves

C_1 and C_2 contained in D and having the same initial and final points.

Thus, if $\vec{F}$ is conservative its line integral $\int_C \vec{F} \cdot d\vec{r}$ is independent of path

Thm: $\int_C \vec{F} \cdot d\vec{r}$ is independent of path in D if and only if $\oint_K \vec{F} \cdot d\vec{r} = 0$ for every closed path K in D

Thm Suppose $\vec{F}$ is a vector field that's continuous on an open connected region D . If $\int_C \vec{F} \cdot d\vec{r}$ is independent of path in D , then $\vec{F}$ is a conservative vector field